

SIMPLE IMPACT STUDIES — SAMPLE DRAFT

Wake County Sample

Traffic Impact Study

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Simple Impact Studies — sample draft

Traffic Impact Study

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EXECUTIVE SUMMARY

This Traffic Impact Analysis (TIA) evaluates the anticipated traffic impacts of the proposed Wake County Sample within Raleigh-Durham CSA, North Carolina (City of Raleigh / Wake County). The analysis follows the structure of the NCDOT Policy on Street and Driveway Access to North Carolina Highways (July 2003) Chapter 5 and the NCDOT Congestion Management Capacity Analysis Guidelines (Standards, March 2022; Best Practices, November 20, 2024). 20 study intersections within a 1.00-mile radius are analyzed under Existing, No-Build, and Build conditions.

20	3	10	6,800
INTERSECTIONS	LOS DROPS	AT LOS E/F	DAILY TRIPS (RAW)

1.0 INTRODUCTION

Project name	Wake County Sample
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	35.8564°, -78.5703°
Build-out (horizon) year	2027
Controlling jurisdiction	City of Raleigh / Wake County
MPO	CAMPO (Capital Area Metropolitan Planning Organization)
Programmed projects source	NCDOT 2026–2035 STIP (adopted July 2025), as amended

1.1 TIA Applicability

Raleigh Street Design Manual Section 7.1.3 triggers a TIA at: 150 or more total peak-hour trips; 100 or more peak-hour trips where primary access is on a two-lane roadway; more than 100 peak-direction trips; or 3,000 or more daily trips. Raleigh UDO Section 8.2.2 (Infrastructure Sufficiency — Streets) governs; Sec. 8.2.2.E.5 requires a traffic mitigation plan evaluated by the Transportation Director.

The proposed development's estimated raw trip generation of 6,800 vehicles per day meets or exceeds the 3,000 vpd threshold at which NCDOT may require a TIA (no reductions are applied in the threshold test).

1.2 Scoping

Effective 7/1/2024, Raleigh DOT scopes the TIA jointly with the applicant team and NCDOT where applicable; reviews are performed by the City's on-call consultant with a fee of \$5,500 per submittal (\$1,000 per addendum). Rezoning TIAs compare maximum current-entitlement trips against maximum proposed-entitlement trips. The background growth factor and the list of approved background developments must be agreed in the Approved Scoping Document prior to TIA submittal; a placeholder for that document is carried in the Appendix.

2.0 METHODOLOGY

NCDOT requires HCM-based analysis at submittal; the Department currently utilizes Synchro 11 (with mandatory SimTraffic simulation, minimum ten runs), HCS for freeway facilities, and Sidra for roundabouts. This screening applies the openly-published Webster/Akçelik signalized model pending those tool runs; Synchro 11-compatible models and SimTraffic outputs are to be provided at formal submittal.

Peak periods	AM and PM weekday peak hours (minimum); counts Tue–Thu, school in session, less than 12 months old
PHF (future conditions)	0.90 per Standards
Heavy vehicles	Average of duals and TTSTs, 2% minimum
RTOR (future analysis)	Not permitted per Standards
Turn-lane storage	Greater of Synchro 95th percentile and SimTraffic maximum queue, rounded up to 25 ft; 100 ft minimum
Roundabout threshold	v/c 0.85
Study-area rule	Intersections where site trips add 10% or more to background volumes on any approach/movement

2.1 Trip Generation Basis

NCDOT's Best Practices call for the ITE Trip Generation Manual with the Department's Rate-vs-Equation spreadsheet, internal capture per the NCHRP 684 spreadsheet method (vehicle occupancy 1.1, 4,000-ft maximum walking distance), and pass-by limited to retail uses with multi-use pass-by capped at 10% of the adjacent street volume. This screening analysis applies public-data trip-generation rates (NHTS 2017 / SANDAG 2002 / NCHRP 716), with every rate tagged so the jurisdiction-approved ITE figure can be substituted at submittal; applied internal-capture and pass-by credits must be reconciled against the NCDOT limits at scoping.

Daily rate applied	80.00 trips per ksf × 85 ksf
AM peak rate applied	3.20 trips per ksf × 85 ksf
PM peak rate applied	8.00 trips per ksf × 85 ksf
Rate source	SANDAG 2002 “(Not So) Brief Guide of Vehicular Traffic Generation Rates for the San Diego Region” — Community Shopping Center (daily 80/ksf; PM pass-by 30%)

3.0 ANALYSIS CONDITIONS AND MEASURES OF EFFECTIVENESS

Scenarios follow the NCDOT Standards set: Existing Base Year; No-Build Design Year (build-out); and Design Year Build. Where a scoped project is phased, intermediate build-out years are analyzed with earlier-phase trips carried as site trips (never background). Where an impacted STIP project is in planning, design, or construction (or within 5 years post-construction) and site traffic was not in the STIP forecast, an additional STIP design-year analysis is coordinated per the Best Practices TIA/STIP flow chart.

3.1 Intersection MOE Summary (NCDOT format)

Approaches are ordered EB, WB, NB, SB and movements left-through-right per the Best Practices reporting format. Signalized intersections report control delay and LOS overall and per lane group (any v/c > 1.0 is reported as LOS F regardless of delay); unsignalized intersections report no overall LOS — results are per lane group with the conflicting movement. SimTraffic maximum-queue columns are populated at submittal.

No.	Intersection	Existing LOS/ Delay	No-Build LOS/ Delay	Build LOS/Delay
1	Fox Rd. & Spring Forest Rd.	C / 25.9s	C / 27.4s	D / 38.9s
2	Hollenden Dr. & Spring Forest Rd.	D / 42.3s	D / 44.6s	E / 73.7s
3	Fox Rd. & Malone Ct. / Sumner Blvd.	C / 26.6s	C / 27.6s	C / 28.4s
4	North New Hope Road & Hollenden Drive	F / 187.8s	F / 196.6s	F / 202.0s
5	Spring Forest Road & Capital Boulevard	F / 300.0s	F / 300.0s	F / 300.0s
6	Fox Road & Louisburg Road	F / 234.6s	F / 244.2s	F / 270.2s
7	Oak Forest Drive & Capital Boulevard	F / 300.0s	F / 300.0s	F / 300.0s

No.	Intersection	Existing LOS/ Delay	No-Build LOS/ Delay	Build LOS/Delay
8	North New Hope Road & Capital Boulevard	D / 37.9s	D / 40.7s	D / 41.8s
9	Capital Boulevard & East Millbrook Road	D / 54.2s	E / 57.8s	E / 60.7s
10	Old Wake Forest & Triangle Towne Blvd	D / 39.1s	D / 42.0s	D / 43.4s
11	Dansey Drive & Louisburg Road	F / 234.6s	F / 244.2s	F / 257.7s
12	Valley Stream Drive & Louisburg Road	F / 234.6s	F / 244.2s	F / 256.6s
13	Spring Forest Road & Louisburg Road	A / 7.9s	A / 7.9s	A / 8.0s
14	Fox Rd. & Old Wake Forest / Segal (walmart)	C / 24.8s	C / 25.9s	C / 26.1s
15	Kyle Drive & Louisburg Road	B / 15.8s	B / 16.0s	B / 16.2s
16	Spring Forest Road & Green Road	A / 9.7s	A / 9.8s	B / 10.1s
17	Louisburg Road & North New Hope Road	F / 188.4s	F / 197.2s	F / 210.5s
18	Sumner Boulevard & Capital Boulevard	F / 248.1s	F / 257.9s	F / 270.8s
19	Capital Boulevard & Calvary Drive	A / 5.2s	A / 5.2s	A / 5.2s
20	Old Wake Forest Road & Town Center Drive	C / 23.4s	C / 23.8s	C / 23.9s

3.2 Trip Distribution — Gravity Model

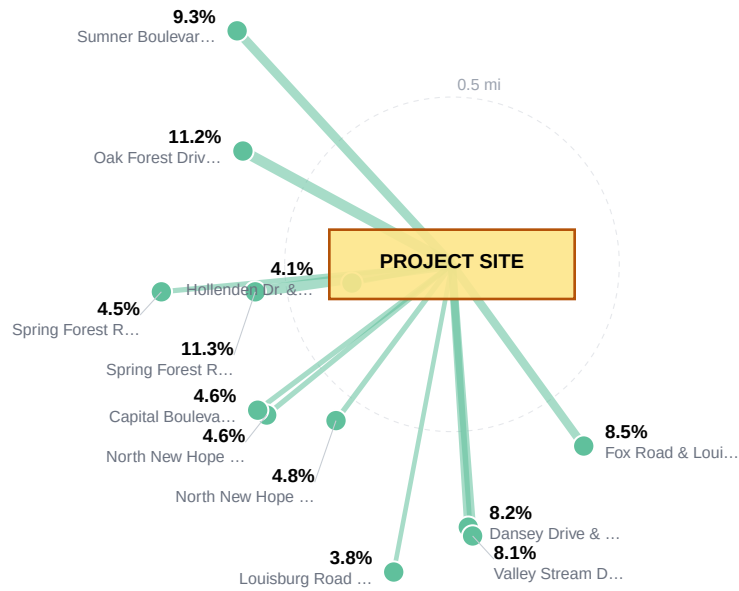
The trip distribution uses the gravity model (term = $M / (d^{1.1} \cdot d_{site})$; mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector		Share of project trips			
Northeast (NNE + ENE)		4%			
Southeast (ESE + SSE)		33%			
Southwest (SSW + WSW)		39%			
Northwest (WNW + NNW)		24%			

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Spring Forest Road & Capital Boulevard	WSW	0.59	5,040	0.1	11.35%
Oak Forest Drive & Capital Boulevard	WNW	0.71	5,220	0.1	11.20%
Sumner Boulevard & Capital Boulevard	NNW	0.95	4,635	0.1	9.29%
Fox Road & Louisburg Road	SSE	0.67	3,825	0.1	8.55%
Dansey Drive & Louisburg Road	SSE	0.79	3,825	0.1	8.22%
Valley Stream Drive & Louisburg Road	SSE	0.81	3,825	0.1	8.15%
North New Hope Road & Hollenden Drive	SSW	0.58	2,070	0.0	4.81%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
North New Hope Road & Capital Boulevard	WSW	0.71	2,070	0.0	4.61%
Capital Boulevard & East Millbrook Road	WSW	0.73	2,070	0.0	4.59%
Spring Forest Road & Green Road	WSW	0.87	2,115	0.0	4.47%
Hollenden Dr. & Spring Forest Rd.	WSW	0.30	1,620	0.0	4.14%
Louisburg Road & North New Hope Road	SSW	0.94	1,845	0.0	3.82%
Fox Rd. & Spring Forest Rd.	ESE	0.20	1,170	0.0	3.11%
Spring Forest Road & Louisburg Road	ESE	0.83	1,170	0.0	2.51%
Kyle Drive & Louisburg Road	ESE	0.86	1,170	0.0	2.48%
Fox Rd. & Old Wake Forest / Segal (walmart)	NNE	0.84	1,000	0.0	2.14%
Old Wake Forest Road & Town Center Drive	NNW	0.98	1,000	0.0	2.04%
Old Wake Forest & Triangle Towne Blvd	NNW	0.77	864	0.0	1.89%
Fox Rd. & Malone Ct. / Sumner Blvd.	NNE	0.51	700	0.0	1.67%
Capital Boulevard & Calvary Drive	SSW	0.97	468	0.0	0.96%

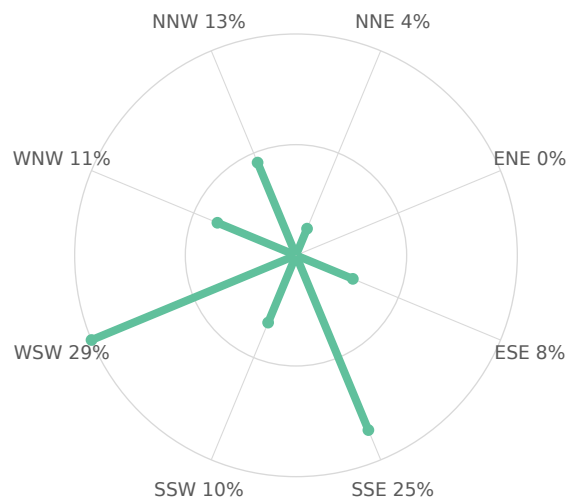
20 study-area zone(s). Screening-grade distribution; not a calibrated regional model.



0.5 mi

Figure — Project Trip Distribution. Study-area zones plotted to scale at their true bearing and distance from the site; leg weight is proportional to each zone's share of project trips, and the label gives that share. Derived from the Gravity Model distribution — the same shares tabulated above. Screening-grade: zone positions are the analysis locations, not a surveyed base map.

Figure — Directional Distribution of Project Trips

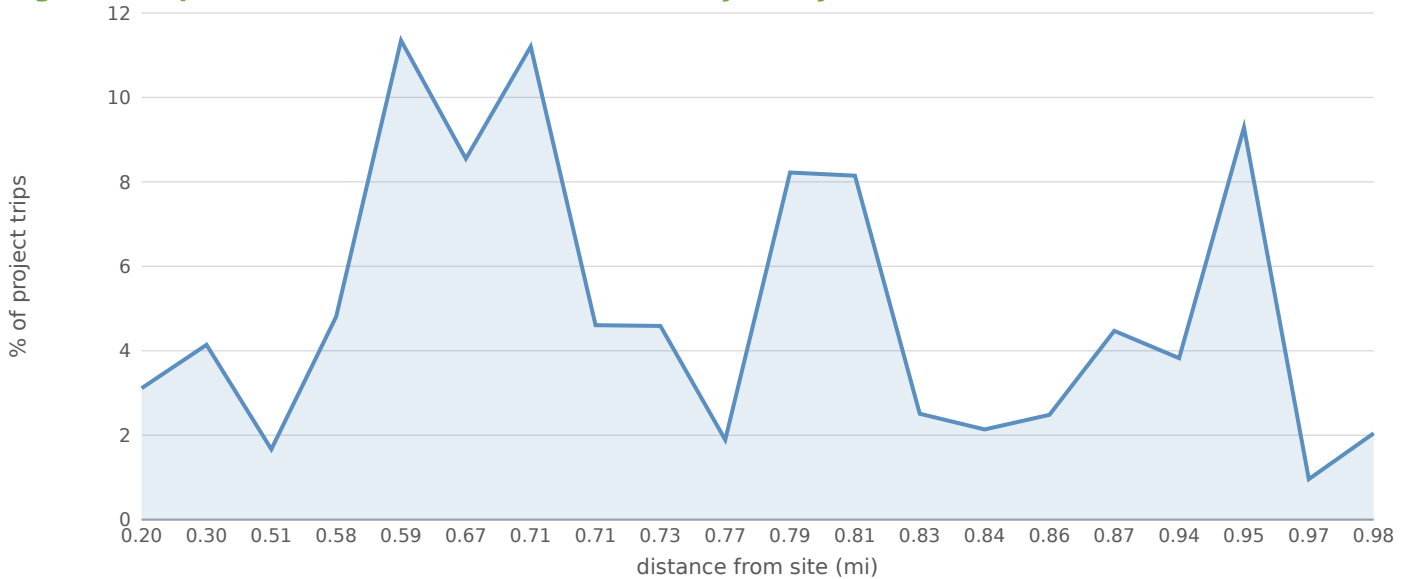


Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone



Figure — Trip Share vs. Distance from Site (Gravity Decay)



3.3 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Fox Rd. & Spring Forest Rd.	ESE	0.20	75	145	13.6%
Hollenden Dr. & Spring Forest Rd.	WSW	0.30	129	240	22.6%
Fox Rd. & Malone Ct. / Sumner Blvd.	NNE	0.51	8	16	1.5%
North New Hope Road & Hollenden Drive	SSW	0.58	12	23	2.2%
Spring Forest Road & Capital Boulevard	WSW	0.59	46	115	10.8%
Fox Road & Louisburg Road	SSE	0.67	56	106	10.0%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Oak Forest Drive & Capital Boulevard	WNW	0.71	21	53	5.0%
North New Hope Road & Capital Boulevard	WSW	0.71	6	12	1.1%
Capital Boulevard & East Millbrook Road	WSW	0.73	8	20	1.9%
Old Wake Forest & Triangle Towne Blvd	NNW	0.77	6	11	1.0%
Dansey Drive & Louisburg Road	SSE	0.79	22	55	5.2%
Valley Stream Drive & Louisburg Road	SSE	0.81	34	51	4.8%
Spring Forest Road & Louisburg Road	ESE	0.83	11	23	2.2%
Fox Rd. & Old Wake Forest / Segal (walmart)	NNE	0.84	2	5	0.5%
Kyle Drive & Louisburg Road	ESE	0.86	7	17	1.6%
Spring Forest Road & Green Road	WSW	0.87	39	58	5.5%
Louisburg Road & North New Hope Road	SSW	0.94	22	55	5.2%
Sumner Boulevard & Capital Boulevard	NNW	0.95	21	53	5.0%
Capital Boulevard & Calvary Drive	SSW	0.97	2	3	0.3%
Old Wake Forest Road & Town Center Drive	NNW	0.98	2	3	0.3%

Project trips assigned to each study intersection from the distribution shares above: a four-step gravity model sets the share reaching each intersection from its attraction mass and its distance from the site. Directional load multipliers are a Florida (Caltran) refinement and are not applied in this region — each intersection's loading is taken directly from the gravity weights above.

4.0 MITIGATION CRITERIA CHECK (POLICY CH. 5.J)

NCDOT requires improvements when, comparing base-network and project conditions: total average delay increases by 25% or more while remaining at the same LOS; the LOS degrades by one level; or the intersection operates at LOS F. Turn-lane mitigation applies where the 95th-percentile queue exceeds existing storage. Signal-timing changes alone are not considered mitigation. The District Engineer makes the final mitigation determination.

No.	Intersection	Δ delay	LOS drop	LOS F	Ch. 5.J result
1	Fox Rd. & Spring Forest Rd.	+42%	YES (C→D)	no	MITIGATION INDICATED
2	Hollenden Dr. & Spring Forest Rd.	+65%	YES (D→E)	no	MITIGATION INDICATED
3	Fox Rd. & Malone Ct. / Sumner Blvd.	+3%	no	no	none indicated
4	North New Hope Road & Hollenden Drive	+3%	no	YES	MITIGATION INDICATED
5	Spring Forest Road & Capital Boulevard	+0%	no	YES	MITIGATION INDICATED
6	Fox Road & Louisburg Road	+11%	no	YES	MITIGATION INDICATED

No.	Intersection	Δ delay	LOS drop	LOS F	Ch. 5.J result
7	Oak Forest Drive & Capital Boulevard	+0%	no	YES	MITIGATION INDICATED
8	North New Hope Road & Capital Boulevard	+3%	no	no	none indicated
9	Capital Boulevard & East Millbrook Road	+5%	no	no	none indicated
10	Old Wake Forest & Triangle Towne Blvd	+3%	no	no	none indicated
11	Dansey Drive & Louisburg Road	+6%	no	YES	MITIGATION INDICATED
12	Valley Stream Drive & Louisburg Road	+5%	no	YES	MITIGATION INDICATED
13	Spring Forest Road & Louisburg Road	+1%	no	no	none indicated
14	Fox Rd. & Old Wake Forest / Segal (walmart)	+1%	no	no	none indicated
15	Kyle Drive & Louisburg Road	+1%	no	no	none indicated
16	Spring Forest Road & Green Road	+3%	YES (A→B)	no	MITIGATION INDICATED
17	Louisburg Road & North New Hope Road	+7%	no	YES	MITIGATION INDICATED
18	Sumner Boulevard & Capital Boulevard	+5%	no	YES	MITIGATION INDICATED
19	Capital Boulevard & Calvary Drive	+0%	no	no	none indicated
20	Old Wake Forest Road & Town Center Drive	+0%	no	no	none indicated

5.0 CONCLUSIONS AND SUBMITTAL NOTES

3 intersections project a LOS drop and 10 operate at LOS E or F under Build conditions; mitigation per Policy Ch. 5.J should be developed with the District Engineer.

Data conventions: NCDOT forecasts supply AADT, % duals and TTSTs, directional split (D), and design-hour factor (K); forecast conversion to peak-hour volumes uses the Intersection Analysis Utility (IAU) in the NCDOT Traffic Engineering Suite. AADT reference: NCDOT ArcGIS AADT mapping application. Crash analysis: a site-radius crash assessment from NCDOT's TEAAS (Traffic Engineering Accident Analysis System) is prepared at submittal — not generated by this screening tool.

This TIA is to be prepared under the direct charge of and sealed by a licensed North Carolina Professional Engineer with expertise in traffic engineering (Policy Ch. 5.A; 21 NCAC 56 .1103 — seal with signature over or adjacent to the seal and date of signing; firm name, address, and NC firm license number on the title block). TIAs marked confidential remain so until a driveway permit is requested or the project is publicly announced.

SCREENING DISCLOSURE: this document is a screening-level draft. It does not include the Approved Scoping Document, SimTraffic outputs, Synchro models, TEAAS crash analysis, or signed/sealed certification — all required for a formal NCDOT TIA submittal.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is a gamma function $F = a \cdot t^b \cdot e^{-(c \cdot t)}$ — the form recommended by NCHRP Report 716 — on the travel time t to the signal, with home-based-work shape parameters $b=-0.02$, $c=-0.123$ (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant a cancels in the share normalization).

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Hollenden Dr. & Spring Forest Rd.	0.30	0.7	240	22.6%
Fox Rd. & Spring Forest Rd.	0.20	0.5	145	13.6%
Spring Forest Road & Capital Boulevard	0.59	1.4	115	10.8%
Fox Road & Louisburg Road	0.67	1.6	106	10.0%
Spring Forest Road & Green Road	0.87	2.1	58	5.5%
Dansey Drive & Louisburg Road	0.79	1.9	55	5.2%
Louisburg Road & North New Hope Road	0.94	2.3	55	5.2%
Oak Forest Drive & Capital Boulevard	0.71	1.7	53	5.0%
Sumner Boulevard & Capital Boulevard	0.95	2.3	53	5.0%
Valley Stream Drive & Louisburg Road	0.81	1.9	51	4.8%
North New Hope Road & Hollenden Drive	0.58	1.4	23	2.2%
Spring Forest Road & Louisburg Road	0.83	2.0	23	2.2%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 87% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(\text{ASC} - \lambda \cdot \Delta \text{GC})})$ calibrated to the metro's auto-mode share — measured (ACS B08301) where the metro is wired, otherwise the ACS B08301 metropolitan median of 90% — and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 13% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 100% of study signals lie on a modeled route; highest loaded-link v/c : 0.37. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Collector	3.24	414	0.37
Minor arterial	2.10	223	0.32
Trunk highway	1.83	84	0.17
Principal arterial	0.04	20	0.16

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 18 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: Σ entering = Σ leaving at 434 network junctions, max imbalance 0.00 veh. 16 study intersection(s) carry path-derived movements; 4 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c , control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Study scope: 20 signalized intersections lie within the 1-mile study area, and all 20 are carried as study intersections. The regional signal inventory holds 52 records inside that radius; 32 were merged into a junction already counted, because a signalized junction is commonly recorded once per approach rather than once per intersection. Merged records are duplicates of an analyzed intersection, not intersections omitted from the study.

Background approach volumes are resolved per leg — the signal's design hour on the main road (its counted volume where the analyzer had a compatible count, else the road-class baseline it assigned — each worksheet says which), the road-class baseline on uncounted legs, client link counts where supplied; half per direction, and half in the physical direction of a one-way carriageway — and the background turning movements in the diagrams are balanced to the exit legs by iterative proportional fitting (NCHRP 255/765 refinement) from a geometry seed; each worksheet states its leg sources and residual. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

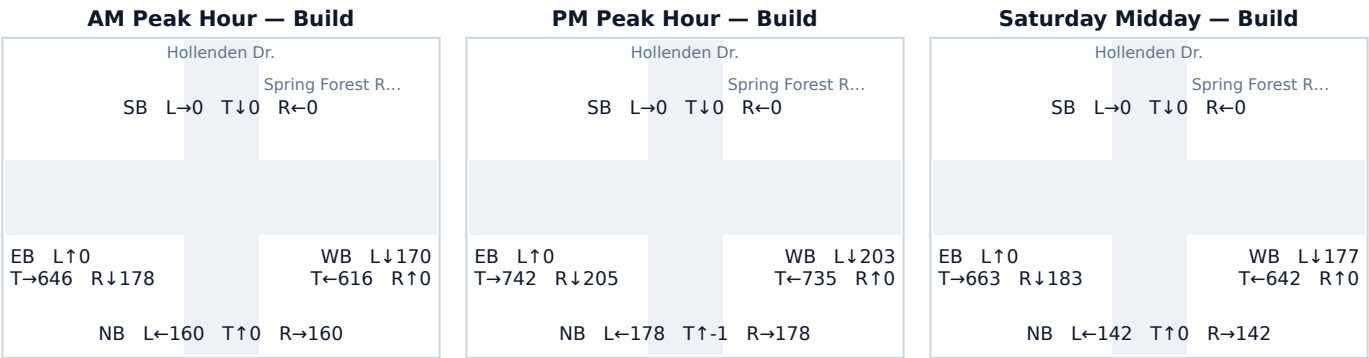
A.1 Hollenden Dr. & Spring Forest Rd.

Signal ID	raleigh-durham-1299179770
Location	SE Raleigh-Durham · 0.30 mi from site
Added PM peak trips	240
Existing / No-Build (PM)	LOS D · 44.6 s/veh · v/c 0.95
Build (PM)	LOS E · 73.7 s/veh · v/c 1.06
Δ control delay	+29.1 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +29.1s, design year +49.4s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 115 s cycle, 3 critical phases; g/C NS 0.23 / EW 0.53 (EW left 0.11); permissive NS / protected EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.0 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	175	175	350
EB approach	175	—	635	810
WB approach	175	635	—	810
Σ exiting	350	810	810	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	0	355	0.87 → 0.87	63.7 → 63.7	E → E	450
SB	0	0	0	0.00 → 0.00	34.3 → 34.3	C → C	0
EB	822	124	947	0.87 → 1.00	34.0 → 55.3	C → E	1,233
WB	822	116	938	0.87 → 0.99	34.0 → 53.1	C → D	1,218

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	124	52%
WB Through	116	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

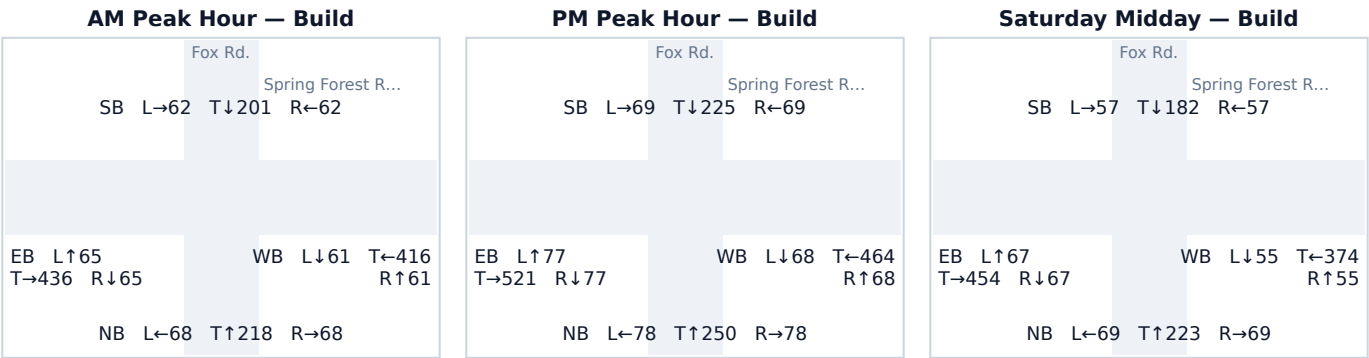
A.2 Fox Rd. & Spring Forest Rd.

Signal ID	raleigh-durham-195601497
Location	SE Raleigh-Durham · 0.20 mi from site
Added PM peak trips	145
Existing / No-Build (PM)	LOS C · 27.4 s/veh · v/c 0.91
Build (PM)	LOS D · 38.9 s/veh · v/c 0.98
Δ control delay	+11.5 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +11.5s, design year +29.7s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.31 / EW 0.52; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 4 iterations, residual 0.0 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	216	67	67	350
SB approach	216	—	67	67	350
EB approach	67	67	—	451	585
WB approach	67	67	451	—	585
Σ exiting	350	350	585	585	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	51	406	0.63 → 0.72	23.0 → 26.2	C → C	248
SB	355	8	363	0.63 → 0.65	23.0 → 23.4	C → C	215
EB	594	80	675	0.63 → 0.72	13.5 → 15.7	B → B	355
WB	594	6	600	0.63 → 0.64	13.5 → 13.6	B → B	296

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Right	55	38%
NB Left	51	35%
EB Through	17	12%
EB Left	8	6%
SB Right	8	6%
WB Through	6	4%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

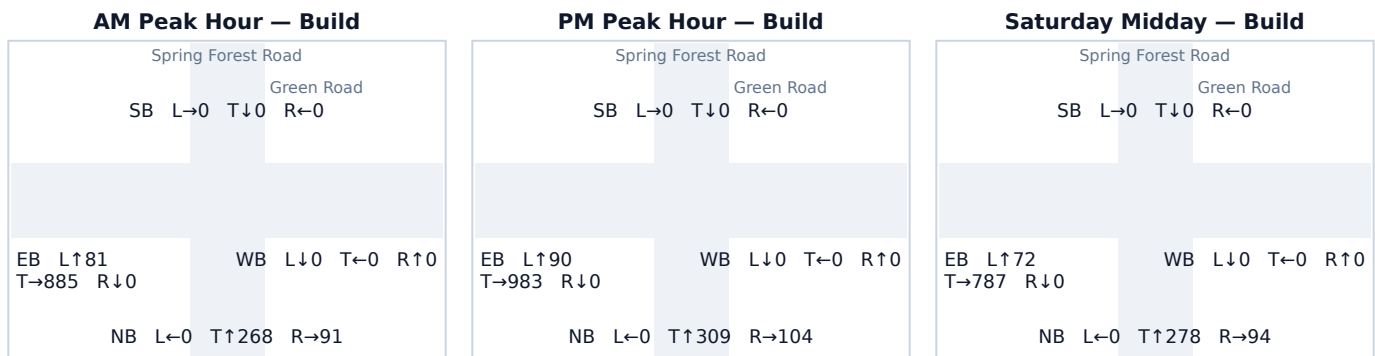
A.3 Spring Forest Road & Green Road

Signal ID	raleigh-durham-6571106896
Location	SE Raleigh-Durham · 0.87 mi from site
Added PM peak trips	58
Existing / No-Build (PM)	LOS A · 9.8 s/veh · v/c 0.53
Build (PM)	LOS B · 10.1 s/veh · v/c 0.55
Δ control delay	+0.3 s/veh (LOS change)
Mitigation	No mitigation indicated at screening level (opening year +0.3s, design year +0.7s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 100 s cycle, 2 critical phases; g/C NS 0.22 / EW 0.68; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 7 iterations, residual 0.3 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	261	0	89	350
SB approach	0	—	0	0	0
EB approach	0	89	—	969	1,058
WB approach	0	0	0	—	0
Σ exiting	0	350	0	1,057	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	58	413	0.88 → 1.03	60.7 → 90.3	E → F	472
SB	0	0	0	0.00 → 0.00	30.1 → 30.1	C → C	0
EB	1,073	0	1,073	0.88 → 0.88	22.4 → 22.4	C → C	987
WB	0	0	0	0.00 → 0.00	5.2 → 5.2	A → A	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Right	58	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

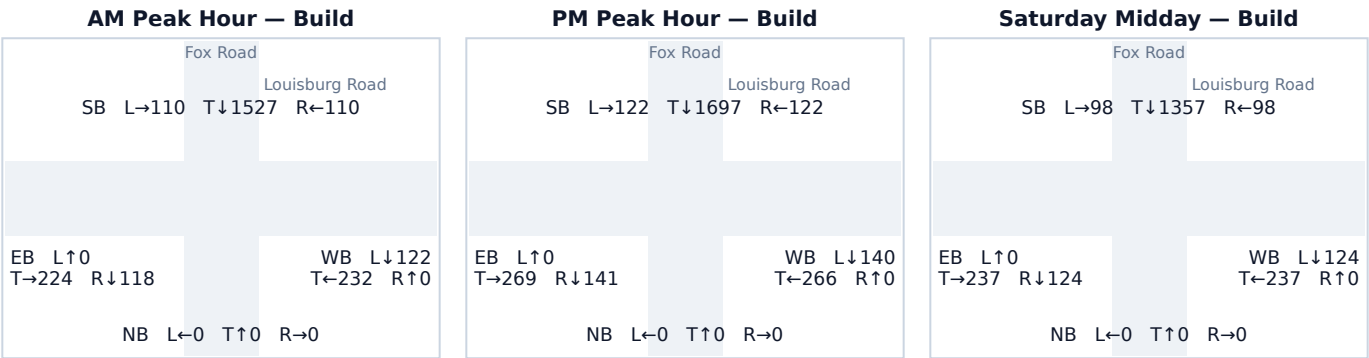
A.4 Fox Road & Louisburg Road

Signal ID	raleigh-durham-6993976381
Location	SE Raleigh-Durham · 0.67 mi from site
Added PM peak trips	106
Existing / No-Build (PM)	LOS F · 244.2 s/veh · v/c 1.47
Build (PM)	LOS F · 270.2 s/veh · v/c 1.53
Δ control delay	+26.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +26.0s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 1.28 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 6 iterations, residual 0.2 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	0	0	0	0
SB approach	1,672	—	121	121	1,913
EB approach	120	0	—	230	350
WB approach	120	0	230	—	350
Σ exiting	1,912	0	350	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
SB	1,941	0	1,941	2.40 → 2.40	300.0 → 300.0	F → F	1,986
EB	355	55	410	0.44 → 0.51	18.7 → 19.9	B → B	301
WB	355	51	406	0.44 → 0.50	18.7 → 19.8	B → B	297

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Right	55	52%
WB Through	51	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

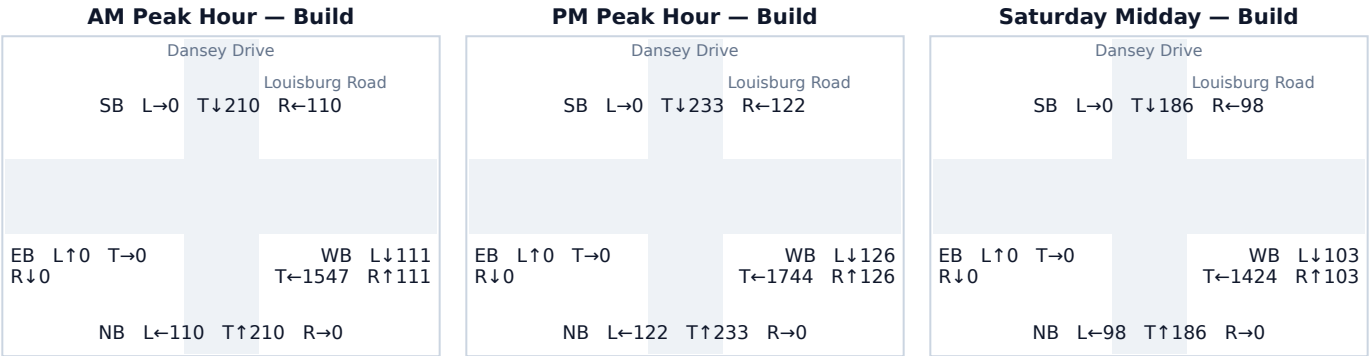
A.5 Dansey Drive & Louisburg Road

Signal ID	raleigh-durham-6569017903
Location	SE Raleigh-Durham · 0.79 mi from site
Added PM peak trips	55
Existing / No-Build (PM)	LOS F · 244.2 s/veh · v/c 1.47
Build (PM)	LOS F · 257.7 s/veh · v/c 1.50
Δ control delay	+13.5 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +13.5s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 1.28 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 6 iterations, residual 0.2 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	230	120	0	350
SB approach	230	—	120	0	350
EB approach	0	0	—	0	0
WB approach	121	121	1,672	—	1,913
Σ exiting	350	350	1,912	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	0	355	0.44 → 0.44	18.7 → 18.7	B → B	251
SB	355	0	355	0.44 → 0.44	18.7 → 18.7	B → B	251
EB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
WB	1,941	55	1,996	2.40 → 2.46	300.0 → 300.0	F → F	2,042

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	55	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

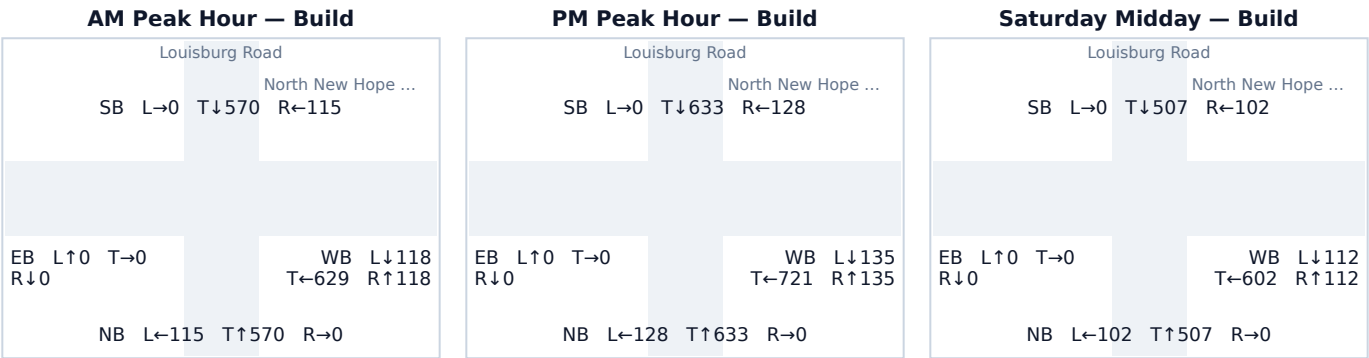
A.6 Louisburg Road & North New Hope Road

Signal ID	raleigh-durham-6569017900
Location	SE Raleigh-Durham · 0.94 mi from site
Added PM peak trips	55
Existing / No-Build (PM)	LOS F · 197.2 s/veh · v/c 1.37
Build (PM)	LOS F · 210.5 s/veh · v/c 1.40
Δ control delay	+13.3 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +13.4s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 1.01 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 5 iterations, residual 0.2 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	624	126	0	750
SB approach	624	—	126	0	750
EB approach	0	0	—	0	0
WB approach	126	126	671	—	923
Σ exiting	750	750	922	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	761	0	761	0.94 → 0.94	43.6 → 43.6	D → D	748
SB	761	0	761	0.94 → 0.94	43.6 → 43.6	D → D	748
EB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
WB	936	55	991	1.16 → 1.22	108.5 → 136.3	F → F	1,014

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Left	36	65%
WB Through	19	35%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

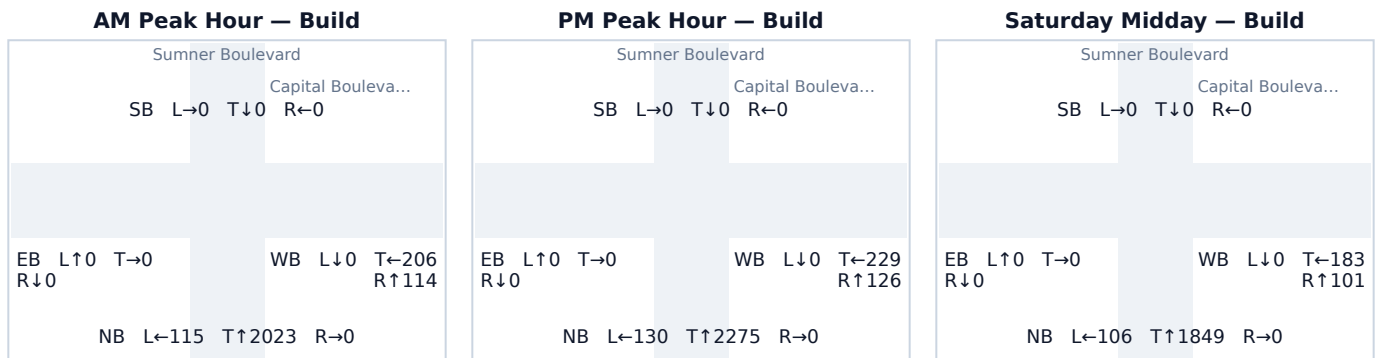
A.7 Sumner Boulevard & Capital Boulevard

Signal ID	raleigh-durham-9401205419
Location	SE Raleigh-Durham · 0.95 mi from site
Added PM peak trips	53
Existing / No-Build (PM)	LOS F · 257.9 s/veh · v/c 1.50
Build (PM)	LOS F · 270.8 s/veh · v/c 1.53
Δ control delay	+12.9 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +12.9s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 1.50 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 7 iterations, residual 0.1 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	2,193	124	0	2,318
SB approach	0	—	0	0	0
EB approach	0	0	—	0	0
WB approach	0	124	226	—	350
Σ exiting	0	2,317	350	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	2,352	53	2,405	2.90 → 2.97	300.0 → 300.0	F → F	2,460
SB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
EB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
WB	355	0	355	0.44 → 0.44	18.7 → 18.7	B → B	251

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	53	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

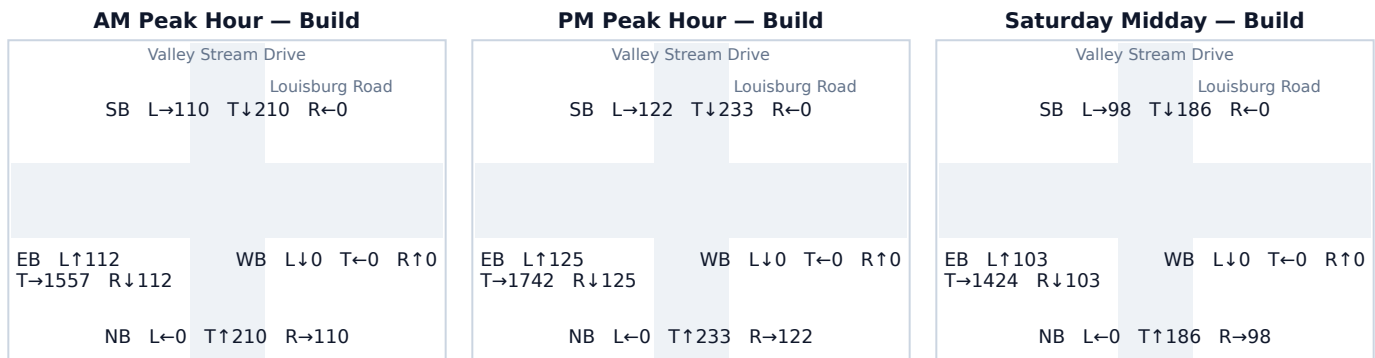
A.8 Valley Stream Drive & Louisburg Road

Signal ID	raleigh-durham-1299218818
Location	SE Raleigh-Durham · 0.81 mi from site
Added PM peak trips	51
Existing / No-Build (PM)	LOS F · 244.2 s/veh · v/c 1.47
Build (PM)	LOS F · 256.6 s/veh · v/c 1.50
Δ control delay	+12.4 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +12.5s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 1.28 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 6 iterations, residual 0.2 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	230	0	120	350
SB approach	230	—	0	120	350
EB approach	121	121	—	1,672	1,913
WB approach	0	0	0	—	0
Σ exiting	350	350	0	1,912	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	0	355	0.44 → 0.44	18.7 → 18.7	B → B	251
SB	355	0	355	0.44 → 0.44	18.7 → 18.7	B → B	251
EB	1,941	51	1,992	2.40 → 2.46	300.0 → 300.0	F → F	2,038
WB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	51	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

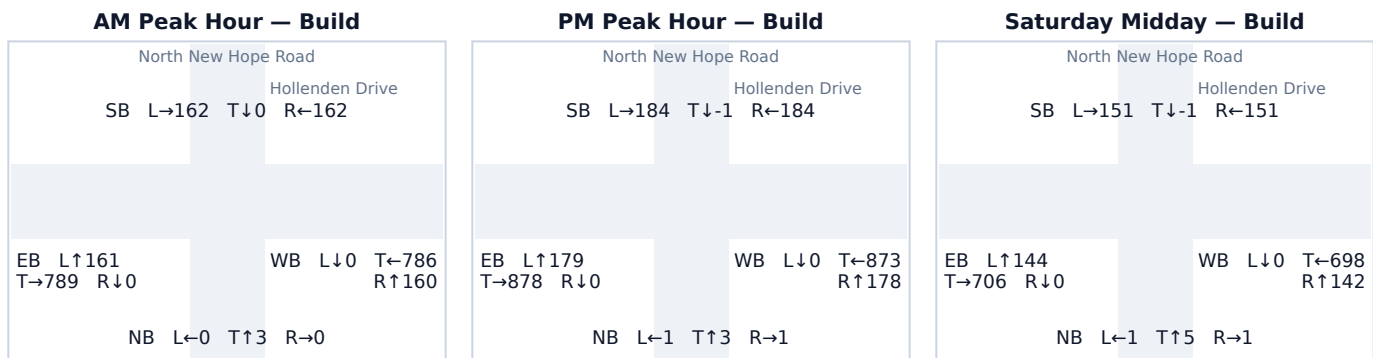
A.9 North New Hope Road & Hollenden Drive

Signal ID	raleigh-durham-1298047232
Location	SE Raleigh-Durham · 0.58 mi from site
Added PM peak trips	23
Existing / No-Build (PM)	LOS F · 196.6 s/veh · v/c 1.36
Build (PM)	LOS F · 202.0 s/veh · v/c 1.38
Δ control delay	+5.4 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +5.5s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 0.88 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 2 iterations, residual 0.0 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ SB leg	→ EB leg	→ WB leg	Σ entering
SB approach	—	175	175	350
EB approach	175	—	860	1,035
WB approach	175	860	—	1,035
Σ exiting	350	1,035	1,035	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	5	5	0.00 → 0.01	13.6 → 13.7	B → B	3

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
SB	355	12	367	0.44 → 0.45	18.7 → 18.9	B → B	262
EB	1,051	6	1,057	1.30 → 1.30	167.3 → 170.5	F → F	1,081
WB	1,051	0	1,051	1.30 → 1.30	167.3 → 167.3	F → F	1,075

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	7	30%
EB Left	6	26%
SB Through	5	22%
NB Through	5	22%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

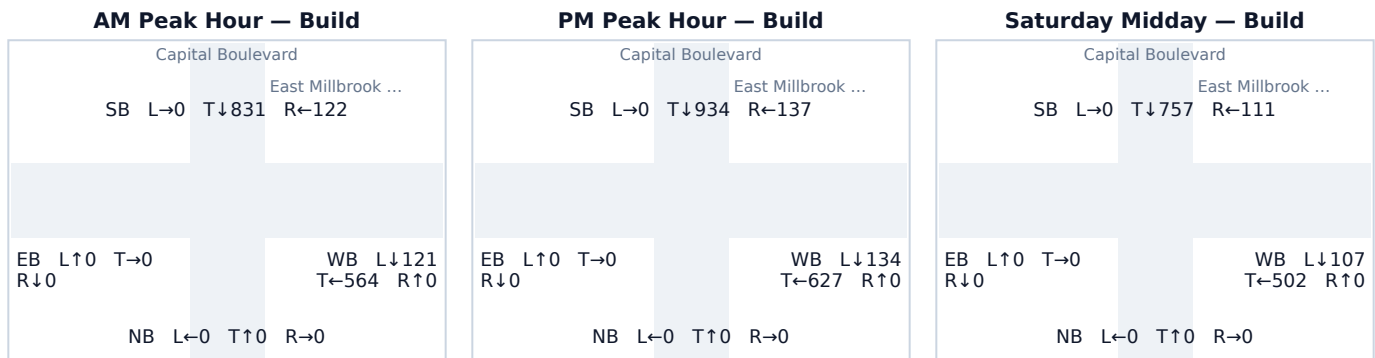
A.10 Capital Boulevard & East Millbrook Road

Signal ID	raleigh-durham-6571199855
Location	SE Raleigh-Durham · 0.73 mi from site
Added PM peak trips	20
Existing / No-Build (PM)	LOS E · 57.8 s/veh · v/c 1.01
Build (PM)	LOS E · 60.7 s/veh · v/c 1.02
Δ control delay	+2.9 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +2.9s, design year +4.8s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 1.01 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 7 iterations, residual 0.3 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	0	0	0	0
SB approach	903	—	132	0	1,035
EB approach	0	0	—	0	0
WB approach	132	0	618	—	750
Σ exiting	1,035	0	750	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
SB	1,051	20	1,071	1.30 → 1.32	167.3 → 177.9	F → F	1,095
EB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
WB	761	0	761	0.94 → 0.94	43.6 → 43.6	D → D	748

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	20	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

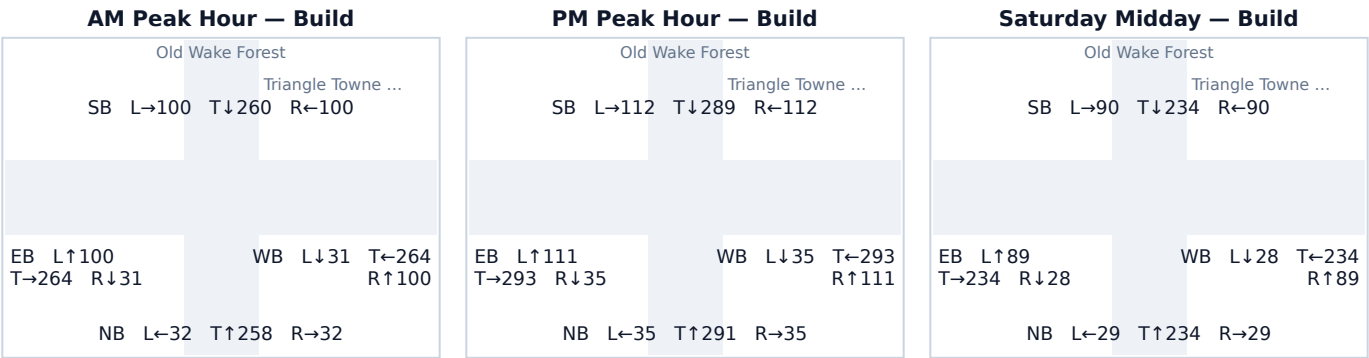
A.11 Old Wake Forest & Triangle Towne Blvd

Signal ID	raleigh-durham-1302112953
Location	SE Raleigh-Durham · 0.77 mi from site
Added PM peak trips	11
Existing / No-Build (PM)	LOS D · 42.0 s/veh · v/c 0.97
Build (PM)	LOS D · 43.4 s/veh · v/c 0.98
Δ control delay	+1.4 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +1.4s, design year +2.7s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.39; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 9 iterations, residual 0.2 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	282	34	34	350
SB approach	282	—	109	109	500
EB approach	34	109	—	289	432
WB approach	34	109	289	—	432
Σ exiting	350	500	432	432	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	6	361	0.44 → 0.45	13.2 → 13.3	B → B	172
SB	508	5	513	0.63 → 0.64	16.5 → 16.7	B → B	273
EB	439	0	439	0.63 → 0.63	19.2 → 19.2	B → B	245
WB	439	0	439	0.63 → 0.63	19.2 → 19.2	B → B	245

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	6	55%
SB Through	5	45%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

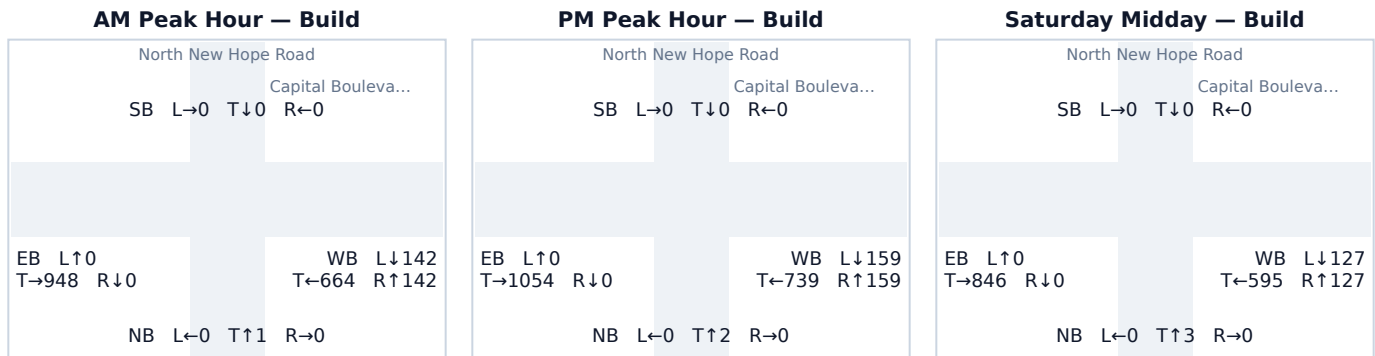
A.12 North New Hope Road & Capital Boulevard

Signal ID	raleigh-durham-10858805422
Location	SE Raleigh-Durham · 0.71 mi from site
Added PM peak trips	12
Existing / No-Build (PM)	LOS D · 40.7 s/veh · v/c 0.97
Build (PM)	LOS D · 41.8 s/veh · v/c 0.97
Δ control delay	+1.1 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +1.1s, design year +2.3s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 85 s cycle, 3 critical phases; g/C NS 0.16 / EW 0.54 (EW left 0.12); permissive NS / protected EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 2 from the signal's counted design hour (half per direction). Turning movements: did not balance within 50 iterations (residual 1035.0 vph; rows held exact); exits scaled to entries, 50% imbalance. 1 extra leg not carried (4×4 matrix). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ EB leg	→ WB leg	Σ entering
EB approach	—	1,035	1,035
WB approach	0	—	0
Σ exiting	0	1,035	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	2	2	0.00 → 0.01	29.8 → 29.8	C → C	2

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
SB	0	0	0	0.00 → 0.00	29.8 → 29.8	C → C	0
EB	1,051	4	1,054	1.08 → 1.08	70.4 → 71.8	E → E	1,015
WB	1,051	6	1,057	1.08 → 1.08	70.4 → 72.6	E → E	1,017

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	4	33%
EB Through	4	33%
WB Left	2	17%
NB Right	2	17%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

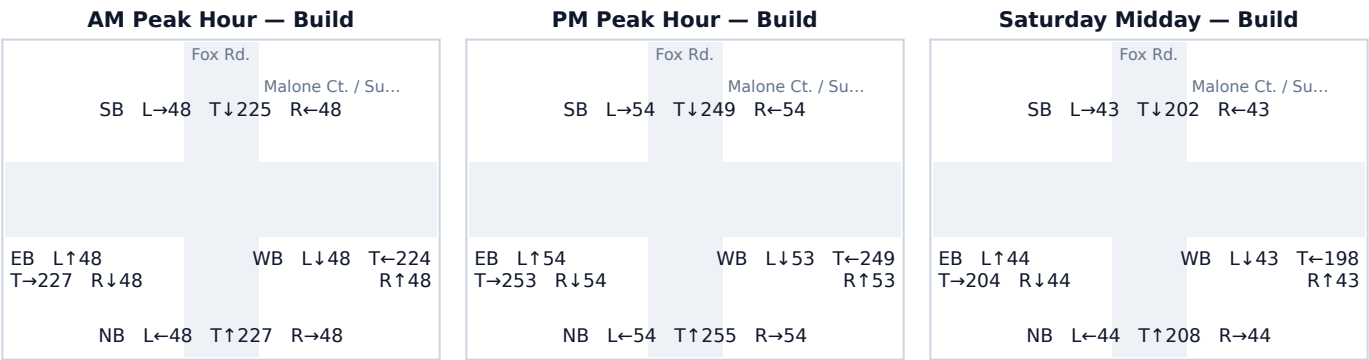
A.13 Fox Rd. & Malone Ct. / Sumner Blvd.

Signal ID	raleigh-durham-1299569010
Location	SE Raleigh-Durham · 0.51 mi from site
Added PM peak trips	16
Existing / No-Build (PM)	LOS C · 27.6 s/veh · v/c 0.85
Build (PM)	LOS C · 28.4 s/veh · v/c 0.86
Δ control delay	+0.8 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +0.8s, design year +3.7s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.42 / EW 0.42; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the analyzer's baseline volume for this signal (no compatible count — road-class or synthetic); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	245	53	53	350
SB approach	245	—	53	53	350
EB approach	53	53	—	245	350
WB approach	53	53	245	—	350
Σ exiting	350	350	350	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	8	363	0.47 → 0.48	14.9 → 15.0	B → B	183
SB	355	2	357	0.47 → 0.48	14.9 → 14.9	B → B	179
EB	355	6	361	0.47 → 0.48	14.9 → 15.0	B → B	181
WB	355	0	355	0.47 → 0.47	14.9 → 14.9	B → B	178

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Left	6	38%
EB Right	6	38%
NB Through	2	13%
SB Through	2	13%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

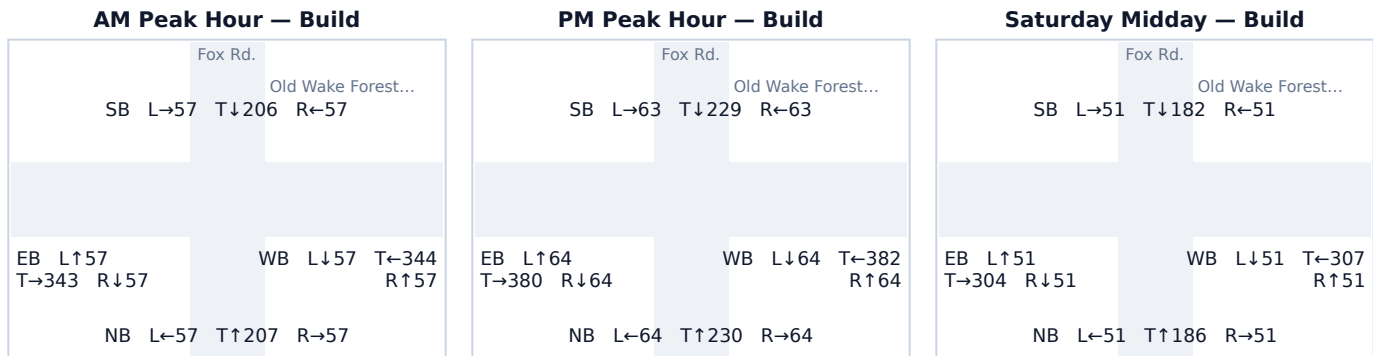
A.14 Fox Rd. & Old Wake Forest / Segal (walmart)

Signal ID	raleigh-durham-195473089
Location	SE Raleigh-Durham · 0.84 mi from site
Added PM peak trips	5
Existing / No-Build (PM)	LOS C · 25.9 s/veh · v/c 0.88
Build (PM)	LOS C · 26.1 s/veh · v/c 0.88
Δ control delay	+0.2 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +0.2s, design year +0.9s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.34 / EW 0.49; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the analyzer's baseline volume for this signal (no compatible count — road-class or synthetic); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.2 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	225	62	62	350
SB approach	225	—	62	62	350
EB approach	62	62	—	375	500
WB approach	62	62	375	—	500
Σ exiting	350	350	500	500	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	3	358	0.58 → 0.58	20.0 → 20.1	B → C	202
SB	355	0	355	0.58 → 0.58	20.0 → 20.0	B → B	200
EB	508	0	508	0.58 → 0.58	13.6 → 13.6	B → B	248
WB	508	2	510	0.58 → 0.58	13.6 → 13.6	B → B	249

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Right	3	60%
WB Left	2	40%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.15 Kyle Drive & Louisburg Road

Signal ID	raleigh-durham-6569017908
Location	SE Raleigh-Durham · 0.86 mi from site
Added PM peak trips	17
Existing / No-Build (PM)	LOS B · 16.0 s/veh · v/c 0.61
Build (PM)	LOS B · 16.2 s/veh · v/c 0.62
Δ control delay	+0.2 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.2s, design year +0.6s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.38; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 16 iterations, residual 0.4 vph). 1 extra leg not carried (4×4 matrix). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	500	85	585
EB approach	0	—	500	500
WB approach	0	0	—	0
Σ exiting	0	500	585	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	594	0	594	0.73 → 0.73	19.4 → 19.4	B → B	336
SB	0	0	0	0.00 → 0.00	9.1 → 9.1	A → A	0
EB	508	17	525	0.73 → 0.76	22.7 → 23.7	C → C	314
WB	0	0	0	0.00 → 0.00	11.4 → 11.4	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	17	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

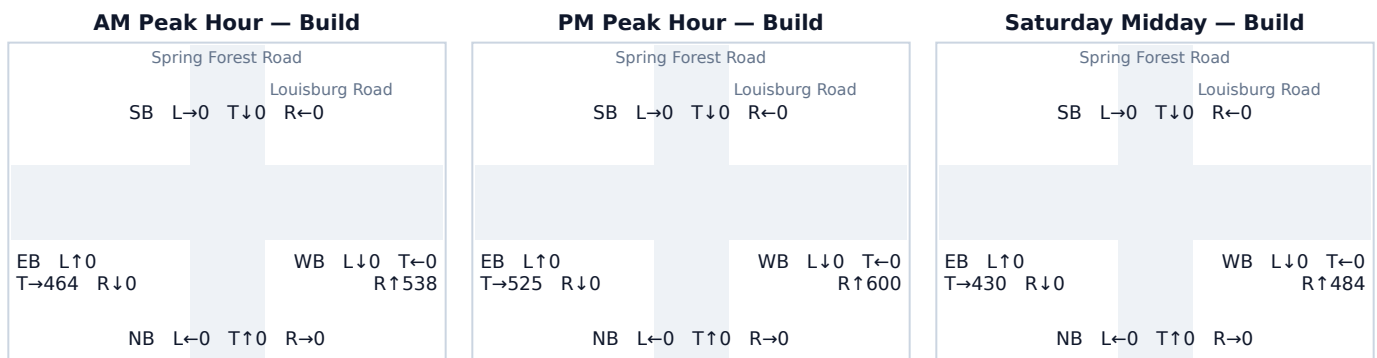
A.16 Spring Forest Road & Louisburg Road

Signal ID	raleigh-durham-1299517371
Location	SE Raleigh-Durham · 0.83 mi from site
Added PM peak trips	23
Existing / No-Build (PM)	LOS A · 7.9 s/veh · v/c 0.46
Build (PM)	LOS A · 8.0 s/veh · v/c 0.47
Δ control delay	+0.1 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.1s, design year +0.2s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.23 / EW 0.60; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: did not balance within 50 iterations (residual 42.5 vph; rows held exact); exits scaled to entries, 8% imbalance. A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ SB leg	→ EB leg	→ WB leg	Σ entering
SB approach	—	0	0	0
EB approach	0	—	500	500
WB approach	585	0	—	585
Σ exiting	585	0	500	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
SB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
EB	508	17	525	0.47 → 0.48	8.1 → 8.2	A → A	203
WB	594	6	600	0.55 → 0.55	9.1 → 9.1	A → A	246

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	17	74%
WB Right	6	26%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

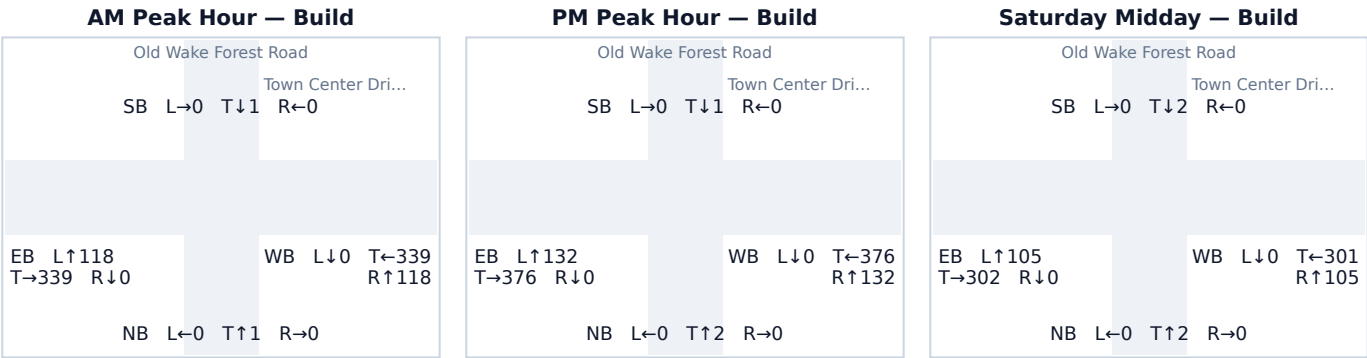
A.17 Old Wake Forest Road & Town Center Drive

Signal ID	raleigh-durham-8521695037
Location	SE Raleigh-Durham · 0.98 mi from site
Added PM peak trips	3
Existing / No-Build (PM)	LOS C · 23.8 s/veh · v/c 0.72
Build (PM)	LOS C · 23.9 s/veh · v/c 0.72
Δ control delay	+0.1 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required. Governing horizon: design year (opening year +0.1s, design year +0.5s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 3 critical phases; g/C NS 0.23 / EW 0.35 (EW left 0.17); permissive NS / protected EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the analyzer's baseline volume for this signal (no compatible count — road-class or synthetic); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph; exits scaled to entries, 35% imbalance). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ SB leg	→ EB leg	→ WB leg	Σ entering
SB approach	—	0	0	0
EB approach	130	—	370	500
WB approach	130	370	—	500
Σ exiting	259	370	370	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	2	2	0.00 → 0.00	17.7 → 17.8	B → B	1
SB	0	1	1	0.00 → 0.00	17.7 → 17.8	B → B	1
EB	508	0	508	0.80 → 0.80	27.7 → 27.8	C → C	315
WB	508	0	508	0.80 → 0.80	27.7 → 27.7	C → C	315

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	1	33%
SB Through	1	33%
NB Left	1	33%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

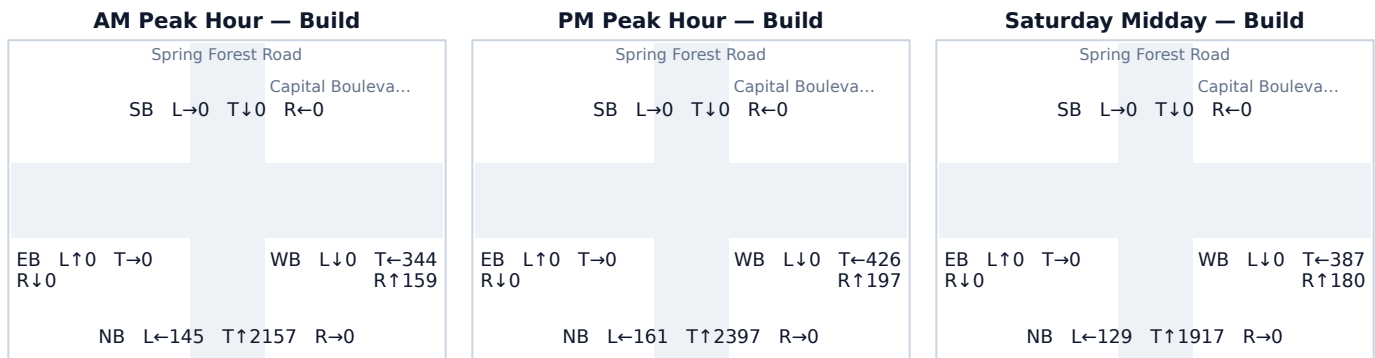
A.18 Spring Forest Road & Capital Boulevard

Signal ID	raleigh-durham-6571106900
Location	SE Raleigh-Durham · 0.59 mi from site
Added PM peak trips	115
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.70
Build (PM)	LOS F · 300.0 s/veh · v/c 1.77
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +0.0s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 1.70 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 7 iterations, residual 0.3 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	2,361	159	0	2,520
SB approach	0	—	0	0	0
EB approach	0	0	—	0	0
WB approach	0	158	342	—	500
Σ exiting	0	2,520	500	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	2,558	0	2,558	3.16 → 3.16	300.0 → 300.0	F → F	2,616
SB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
EB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
WB	508	115	623	0.63 → 0.77	22.6 → 27.8	C → C	540

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	63	55%
WB Right	52	45%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

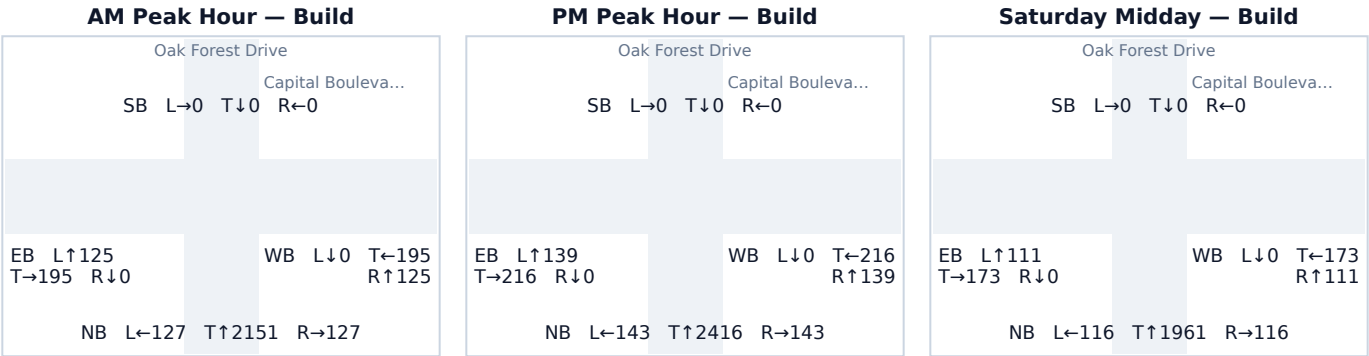
A.19 Oak Forest Drive & Capital Boulevard

Signal ID	raleigh-durham-6561885220
Location	SE Raleigh-Durham · 0.71 mi from site
Added PM peak trips	53
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.87
Build (PM)	LOS F · 300.0 s/veh · v/c 1.90
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +0.0s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 1.67 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 6 iterations, residual 0.1 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	2,336	137	137	2,610
SB approach	0	—	0	0	0
EB approach	0	137	—	213	350
WB approach	0	137	213	—	350
Σ exiting	0	2,610	350	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	2,649	53	2,702	3.27 → 3.34	300.0 → 300.0	F → F	2,764
SB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
EB	355	0	355	0.44 → 0.44	18.7 → 18.7	B → B	251
WB	355	0	355	0.44 → 0.44	18.7 → 18.7	B → B	251

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	53	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

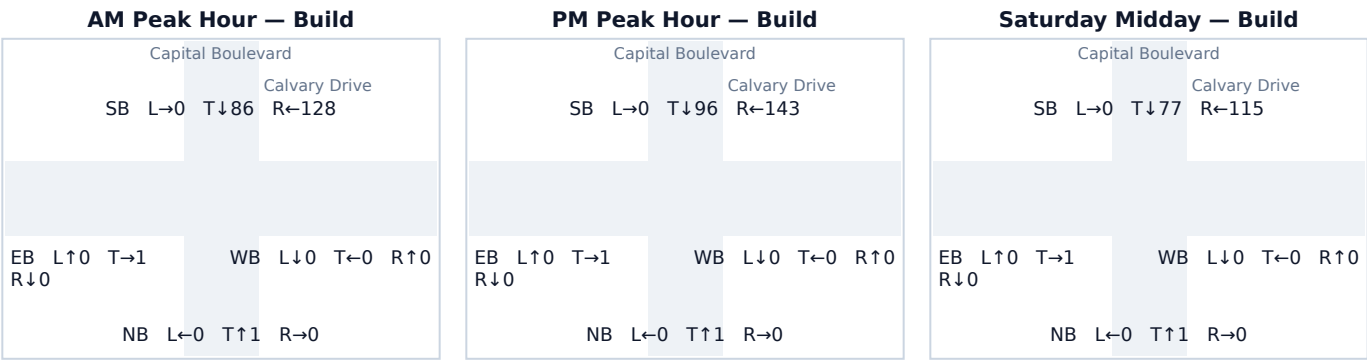
A.20 Capital Boulevard & Calvary Drive

Signal ID	raleigh-durham-6571199856
Location	SE Raleigh-Durham · 0.97 mi from site
Added PM peak trips	3
Existing / No-Build (PM)	LOS A · 5.2 s/veh · v/c 0.10
Build (PM)	LOS A · 5.2 s/veh · v/c 0.10
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.60 / EW 0.23; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph; exits scaled to entries, 150% imbalance). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	Σ entering
NB approach	—	0	0	0
SB approach	94	—	140	234
EB approach	0	0	—	0
Σ exiting	94	0	140	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	1	0.00 → 0.00	4.7 → 4.7	A → A	0
SB	238	2	239	0.22 → 0.22	5.9 → 5.9	A → A	75
EB	0	1	1	0.00 → 0.00	17.7 → 17.8	B → B	0
WB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	1	33%
EB Left	1	33%
SB Through	1	33%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.